Postulates of Quantum Mechanics

Quantum Physics, 2nd Exercise Sheet – Solutions

Exercises

Exercise 1: Classical master equation

Consider a two-state classical system with probabilities $p_0(t)$ and $p_1(t)$. The dynamics is given by the master equation

$$\frac{d}{dt}\begin{pmatrix} p_0 \\ p_1 \end{pmatrix} = \begin{pmatrix} -\alpha & \beta \\ \alpha & -\beta \end{pmatrix}\begin{pmatrix} p_0 \\ p_1 \end{pmatrix}.$$

a) Show that the system has a stationary solution and find it. */Check whether the rate matrix has a zero eigenvalue, and find the corresponding eigenvector./*

Solution. The stationary state $\mathbf{p}^{\mathrm{st}}$ satisfies $M\mathbf{p}^{\mathrm{st}} = 0$, where

$$M = \begin{pmatrix} -\alpha & \beta \\ \alpha & -\beta \end{pmatrix}.$$

I.e. if M has a zero eigenvalue, it has a stationary solution: $\det M = \alpha\beta - \alpha\beta = 0$, so it has a stationary solution.

Solving the eigenvector equaion gives $\alpha p_0^{\mathrm{st}} = \beta p_1^{\mathrm{st}}$ and $p_0^{\mathrm{st}} + p_1^{\mathrm{st}} = 1$. Hence:

$$\boxed{p_0^{\mathrm{st}} = \frac{\beta}{\alpha+\beta}}, \quad \boxed{p_1^{\mathrm{st}} = \frac{\alpha}{\alpha+\beta}}.$$

□

b) Find the non-zero eigenvalue and the corresponding eigenvector of the rate matrix. Find the "solution" when the system is initialized in this normalized eigenvector.

Solution.

$$\det(M - \lambda I) = (\alpha+\lambda)(\beta+\lambda) - \alpha\beta = \lambda(\lambda+\alpha+\beta) = 0 \implies \lambda = -(\alpha+\beta).$$

The eigenvector is the solution of the equation:

$$\begin{pmatrix} -\alpha & \beta \\ \alpha & -\beta \end{pmatrix}\begin{pmatrix} 1 \\ a \end{pmatrix} = -(\alpha+\beta)\begin{pmatrix} 1 \\ a \end{pmatrix} \implies a = -1$$

A normalized form is $\mathbf{p}^t = \frac{1}{\sqrt{2}}(1,-1)^T$. If initialized in this "mode":

$$\frac{d}{dt}\mathbf{p}^t(t) = M\mathbf{p}^t(t) = -(\alpha+\beta)\mathbf{p}^t(t) \implies \mathbf{p}^t(t) = e^{-(\alpha+\beta)t}\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

which decays exponentially to zero. □

c) Write up the general solution of the system with any initial condition as the linear combination of the stationary state and the other, transient part of the state.

Solution. The general solution is

$$\mathbf{p}(t) = a\mathbf{p}_{\mathrm{st}} + b\,e^{-(\alpha+\beta)t}\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

However, $p_0(t) + p_1(t) = 1$, which implies that $a = 1$. The b coefficient is determined by the initial condition:

$$b = \frac{p_0(0) - \beta/(\alpha+\beta)}{1}.$$

This guarantees conservation of total probability and exponential relaxation to the stationary state. □

Exercise 2: Pauli matrices and Hermitian operators

Any 2×2 Hermitian operator can be written as

$$H = a_0 I + \mathbf{a} \cdot \boldsymbol{\sigma}, \quad \mathbf{a} \in \mathbb{R}^3,$$

where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices.

a) Show that the eigenvalues of H are $\lambda_\pm = a_0 \pm |\mathbf{a}|$. */Hint: Write H as a 2-by-2 matrix and calculate its eigenvalues explicitly, using the characteristic polynomial* $\det(H - \lambda I) = 0$./

Solution.

$$H = \begin{pmatrix} a_0 + a_z & a_x - i a_y \\ a_x + i a_y & a_0 - a_z \end{pmatrix} \implies \det(H - \lambda I) = (a_0 + a_z - \lambda)(a_0 - a_z - \lambda) - a_x^2 - a_y^2 = 0$$

$$(a_0 + a_z - \lambda)(a_0 - a_z - \lambda) - a_x^2 - a_y^2 = \lambda^2 - 2\lambda a_0 + a_0^2 - a_z^2 - a_x^2 - a_y^2 = (\lambda - a_0)^2 - |\mathbf{a}|^2 = 0 \implies \lambda_\pm = a_0 \pm |\mathbf{a}|$$

□

b) Show that for any two vectors $(\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma}) = \mathbf{a} \cdot \mathbf{b} I + i(\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma}$. */Use index notations, and the fact that* $\sigma_j \sigma_k = \delta_{jk} I + i \sum_l \epsilon_{jkl} \sigma_l$./

Solution.

$$(\mathbf{a}\cdot\boldsymbol{\sigma})(\mathbf{b}\cdot\boldsymbol{\sigma}) = \sum_{jk} a_j b_k \sigma_j \sigma_k = \sum_{jk} a_j b_k (\delta_{jk} I + i \sum_l \epsilon_{jkl} \sigma_l) = \sum_{jk} a_j b_k \delta_{jk} I + i \sum_{jkl} \epsilon_{ljk} a_j b_k \sigma_l (\mathbf{a}\cdot\mathbf{b}) I + i(\mathbf{a}\times\mathbf{b})\cdot\boldsymbol{\sigma}.$$

□

c)* Show that

$$e^{-iHt} = e^{-ia_0 t} \left(\cos(|\mathbf{a}|t)\, I - i \sin(|\mathbf{a}|t)\, \hat{\mathbf{a}} \cdot \boldsymbol{\sigma} \right),$$

using the Taylor series of the exponential and trigonometric functions.

Solution. Since I and $\mathbf{a} \cdot \boldsymbol{\sigma}$ commute, expand:

$$e^{-iHt} = e^{-ia_0 t} e^{-i(\mathbf{a}\cdot\boldsymbol{\sigma})t}.$$

Using the series and $(\mathbf{a} \cdot \boldsymbol{\sigma})^2 = |\mathbf{a}|^2 I$ and separating even and odd terms:

$$e^{-i(\mathbf{a}\cdot\boldsymbol{\sigma})t} = \sum_{n=0}^{\infty} \frac{(-it)^n (\mathbf{a}\cdot\boldsymbol{\sigma})^n}{n!} = \cos(|\mathbf{a}|t) I - i \sin(|\mathbf{a}|t) \hat{\mathbf{a}} \cdot \boldsymbol{\sigma}.$$

Combining terms gives the desired expression. □

Exercise 3: Expectation values and uncertainty

For the three quantum states

$$|0\rangle, \quad |+\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |\phi\rangle = \tfrac{1}{2}(|0\rangle + \sqrt{3}\,|1\rangle),$$

compute:

a) The expectation values $\langle\sigma_x\rangle_\psi$, $\langle\sigma_y\rangle_\psi$, $\langle\sigma_z\rangle_\psi$.

Solution. For $|0\rangle$: $\langle\sigma_x\rangle = 0$, $\langle\sigma_y\rangle = 0$, $\langle\sigma_z\rangle = 1$. For $|+\rangle$: $\langle\sigma_x\rangle = 1$, $\langle\sigma_y\rangle = 0$, $\langle\sigma_z\rangle = 0$. For $|\phi\rangle$: $\langle\sigma_x\rangle = \frac{\sqrt{3}}{2}$, $\langle\sigma_y\rangle = 0$, $\langle\sigma_z\rangle = -\frac{1}{2}$. □

b) The variances $(\Delta_\psi \sigma_x)^2$, $(\Delta_\psi \sigma_y)^2$, $(\Delta_\psi \sigma_z)^2$.

Solution. Since $\sigma_i^2 = I$, $(\Delta_\psi \sigma_i)^2 = 1 - \langle\sigma_i\rangle^2$. Thus:

$$(\Delta\sigma_x)^2 = 1 - \langle\sigma_x\rangle^2, \quad (\Delta\sigma_y)^2 = 1 - \langle\sigma_y\rangle^2, \quad (\Delta\sigma_z)^2 = 1 - \langle\sigma_z\rangle^2.$$

Plugging in the values gives:

State	$(\Delta\sigma_x)^2$	$(\Delta\sigma_y)^2$	$(\Delta\sigma_z)^2$
$\|0\rangle$	1	1	0
$\|+\rangle$	0	1	1
$\|\phi\rangle$	$1 - \frac{3}{4} = \frac{1}{4}$	1	$1 - \frac{1}{4} = \frac{3}{4}$

□

c) Demonstrate the Heisenberg uncertainty relation for all states:

$$\Delta_\psi \sigma_y \, \Delta_\psi \sigma_z \geq \frac{|\langle\psi| [\sigma_y, \sigma_z] |\psi\rangle|}{2}.$$

Solution. Notice that $[\sigma_y, \sigma_z] = 2i\sigma_x$. The left and right hand sides of the equation, for the three different states, collected in a table:

State	LHS $(\Delta_\psi \sigma_y \, \Delta_\psi \sigma_z)$	RHS $\left(\frac{|\langle\psi|[\sigma_y,\sigma_z]|\psi\rangle|}{2}\right)$
$\|0\rangle$	0	0
$\|+\rangle$	1	1
$\|\phi\rangle$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{2}$

. (1)

Thus, the inequality is satisfied for all three states, and it is saturated for all cases. □

Exercise 4: Hadamard gate

The Hadamard gate is defined by

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

a) Verify that $H^2 = I$ and that H is unitary.

Solution.

$$H^2 = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = I.$$

Since $H^\dagger = H$ and $H^\dagger H = H^2 = I$, H is unitary. □

b) Determine the global phase of the Hadamard gate from its determinant: $e^{i\alpha} = \det H$.

Solution.

$$\det H = \frac{1}{2}(1 \cdot (-1) - 1 \cdot 1) = -1 = e^{i\pi},$$

hence $\boxed{\alpha = \pi}$. □

c) Determine the rotation axis $\hat{\mathbf{n}}$ and angle θ such that

$$H = e^{i\frac{\alpha}{2}} e^{-i\frac{\theta}{2}\hat{\mathbf{n}}\cdot\boldsymbol{\sigma}}.$$

/Hint: Use the Pauli decomposition of the matrix $e^{-i\frac{\alpha}{2}} H = e^{-i\frac{\theta}{2}\hat{\mathbf{n}}\cdot\boldsymbol{\sigma}}$ and the identity in Exercise 2c)./

Solution.

$$e^{-i\frac{\theta}{2}\hat{n}\cdot\boldsymbol{\sigma}} = \frac{e^{-i\pi/2}}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -i & -i \\ -i & i \end{pmatrix} = -i\frac{1}{\sqrt{2}}(\sigma_z + \sigma_x) \implies \cos(\theta/2) = 0, \sin(\theta/2) = 1.$$

Thus $\boxed{\theta = \pi}$ and $\boxed{\hat{\mathbf{n}} = (\hat{x} + \hat{z})/\sqrt{2}}$. □